

Hanbyul Choi

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Areas of Interest

Biomedical Informatics / Interpretable Machine Learning Architecture / Survival Analysis

Education

Yonsei University

Wonju, Republic of Korea

B.S., Division of Software (Double Major in Medical AI and Semiconductor)

Relevant Coursework: Machine Learning, Algorithms, Linear Algebra, Medical Data Analysis

GPA [3.97 / 4.5]

Mar 2022 – Feb 2027 (Expected)

University of California, San Diego

California, United States

Department of Computer Science and Engineering

Exchange Student

Jan 2026 – Jun 2026

Grenoble INP Ensimag, University of Grenoble Alpes

Grenoble, France

AI Summer School

Jul 2024 – Jul 2024

Publication

Choi, H., Sung, S., Jeong, S. W., Choi, H. J., & Kim, M.

A Risk-Stratified Mixture-of-Experts Architecture for Explainable Survival Analysis in Geriatric Lung Cancer (In preparation)

Dec 2025 – Present

- Designed a Mixture-of-Experts architecture for survival analysis, prioritizing calibration and interpretability rather than discrimination alone.
- First author; led conceptualization, research design, algorithmic implementation, and manuscript preparation.
- Selected and funded by the Brain Korea 21 (BK21) Interdisciplinary Research Program.

Research Experience

Yonsei University Wonju College of Medicine

Wonju, Republic of Korea

Undergraduate AI Research Intern (Advised by Prof. Misun Kim)

Jul 2025 – Present

- Analyzed medical data characteristics and utilized hospital software systems, including PACS and MIM, for clinical data processing.
- Conducted research on identifying key biomarkers in lung cancer patients using PACS, MIM, and artificial intelligence techniques.
- Performed AI-based automated segmentation and identified statistically significant association between sarcopenia and survival outcomes in geriatric lung cancer patients (log-rank $p < 0.05$).

Leadership and Activities

Yonsei Murray Volunteer Association
SWeeT Team (Software Education Volunteer)

Wonju, Republic of Korea
Jan 2024

- Conducted a one-week educational program at a local children's center in Pohang on the Fourth Industrial Revolution and software technologies.

Yonsei University Badminton Club SNAL
President

Wonju, Republic of Korea
Jan 2023 – Dec 2023

- Revitalized a student organization that had declined due to the COVID-19 pandemic by implementing structural reforms and organizing multiple events, including the first Homecoming Day in the club's history.

Scholarships

Scholarship Student of the 4th Korea-U.S. Student Exchange Program in the field of high-tech industry
Ministry of Trade, Industry and Resources 2025

Merit-based Scholarship
Yonsei University SW-Centered University Program 2022 Fall, 2024 Spring

Merit-based Scholarship
Yonsei University, Wonju 2023 Spring, 2026 Spring

Awards

Yonsei University 5th Medical AI Solution Coding Competition
4th place 2025

Yonsei University 4th Medical AI Solution Coding Competition
3rd place 2025

Skills

Technical: C/C++, Python, PyTorch, Scikit-learn

Language: Korean: Native, English: Fluent (TOEFL 88, Exchange student at UCSD)